These values are known as 'Atwater Bryant factors' or physiological fuel values.

9.7.6 Coefficient of digestibility

The coefficient of digestibility is used to express the proportion of an ingested nutrient that ultimately becomes available to the body cells. The coefficient of digestibility for carbohydrate, fat and protein are 0.98, 0.95 and 0.92 respectively. It is observed that carbohydrate and fat are metabolized almost completely, whereas protein metabolism is incomplete due to the presence of nitrogen.

The physiological fuel value, Co-efficient of digestibility and digestibility percent of carbohydrate, fat and proteins is presented in **table 9.4**.

9.8 Basal metabolism

Basal Metabolism is the minimum amount of energy needed by the body for maintenance of life when the person is at complete physical and mental rest and having normal body temperature and in the post -absorptive state (12 hours after the intake of last meal). Basal Metabolic Rate (BMR) is a measure of the energy required by the activities of resting tissue.

The Basal Metabolic rate can be measured directly from the heat produced (using a Respiration Calorimeter and Metabolic Chamber) or indirectly from O_2 intake and CO_2 expenditure when the subject is at rest.

9.8.1 Basal Metabolic Rate (BMR)

GRBCM9

The rate of energy use for metabolism under basal metabolism is usually expressed as kcal/kg body weight per hour. The factors affecting BMR is presented in **table 9.5**

ACTIVITY - 3

List the energy , protein and fat value of any 4 commonly consumed foods (per 100g):

a) cereals

b) fruits

c) nuts

d) dairy products

Table 9.4 Types and Sources of Carbohydrates

Nutrient	Heat of combustion (kCal)	Co-efficient of digestibility	Digestibility percent	Physiological fuel value
Carbohydrate	4.1	0.98	98	4.0
Fat	9.45	0.95	95	9.0
Protein	5.65	0.92	92	4.0

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